ROAD BIKES · 5 MIN READ

Frame material, brakes and tyres

Aluminium or carbon

- **Aluminium** is tough and costs less. It is a little heavier and passes more road vibration to the rider. Most aluminium road bikes come with a carbon fork to soften the front.
- **Carbon** is lighter. The maker can make it stiff where power goes in and softer where comfort matters. It costs more and needs care: tighten bolts with a torque wrench and have the frame checked after a crash.

Reading the model name

BRAND	HOW TO READ IT
Trek	**AL** is aluminium. **SL** is carbon. **SLR** is Trek's top carbon.
Merida	Numbers in the hundreds (100 to 400) are aluminium. Numbers in the thousands (4000 and up) are carbon.
Cannondale	**CAAD** is aluminium. SuperSix EVO is always carbon. **LAB71** is the top tier.

Examples from our site: Merida Scultura Endurance 300 is aluminium, Scultura Endurance 4000 is carbon. Trek Domane AL 5 is aluminium, Domane SL 5 is carbon.

Disc or rim brakes

- **Disc brakes** stop the same way in rain and dry, which matters in monsoon. They also leave room for wider tyres. Almost every new road bike is disc.
- **Hydraulic** discs use fluid: light lever feel and strong braking. **Mechanical** discs use a cable: cheaper, more hand effort.
- **Rim brakes** are lighter and cheaper. We still list some, like the Merida Scultura Rim 100. Rim brake wheels and disc wheels cannot be swapped between bikes.

Tyres

28 to 32 mm is normal on a road bike now. A wider tyre at lower pressure gives more comfort and grip, and on real roads it is not slower. Endurance bikes take the widest tyres: the Trek Domane clears up to 38 mm.

Remember these

- Trek AL is aluminium, SL and SLR are carbon.
- Merida hundreds are aluminium, thousands are carbon.
- Disc brakes work the same in rain and allow wider tyres.
- Rim brake wheels do not fit disc brake bikes.